ECE 370 LAB 2

Introduction to AVR Simulation with Atmel Studio

Lab session: 011

Time: 12:00-1:50pm

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Introduction

Lab 2 involves getting familiarized with Atmel Studio's debugging mode. The process has us set break points in a sample code and read different pointer values, register values, and manipulate memory values.

Program Overview

The sample program given initializes the stack pointer and different register values, running through a few loops to initialize some of the registers. It then runs through a function call that generates values into the $0104 and $0105 memory locations

Study Questions

1. What is the initial value of DDRB ?

The initial value of DDRB is 0x00 (hexadecimal) which is 0 in decimal.

2. What is the initial value of PORTB ?

The initial value of PORTB is also 0x00, which equates to 0.

3. Based on the initial values of DDRB and PORTB, what is Port B's default I/O configuration ?

Based on the initial values of DDRB and PORT B, Port B's default I/O configuration is set for output.

4. What 16-bit address (in hexadecimal) is the stack pointer initialized to ?

The stack pointer is initialized to 0x00FF (hexadecimal), which is 255 in decimal.

5. What are the contents of register r0 after it is initialized ?

The contents of register r0 is 0xFF after it is initialized, which is 255 in decimal.

6. How many times did the code inside of LOOP end up running ?

The loop ends up running 4 times.

7. Which instruction would you modify if you wanted to change the number of times that the loop runs ?

To change the number of times that the loop runs, I would change the line of "ldi i, $04" and adjust the number followed by the dollar sign.

8. What are the contents of register r1 after it is initialized ?

The contents of r1 = 0xAA.

9. What are the contents of register r2 after it is initialized ?

The contents of r2 = 0x0F

10. What are the contents of register r3 after it is initialized ?

The contents of r3 = 0x0F.

11. What is the value of the stack pointer when the program execution is inside the FUNCTION subroutine ?

The value of the stack pointer is 0x10FD.

12. What is the final result of FUNCTION? (What are the hexadecimal contents of memory locations $0105:$0104) ?

The final result of FUNCTION is $0104 = 0e and $0105 = ba.

DIFFICULTIES

Some difficulties we encountered were initially finding which windows to open to find the correct values for specific registers. Another difficulty was initially finding the specific locations within the windows to find the values for the registers. General difficulties would be getting familiar with the debugging process.

CONCLUSION

In conclusion, through this lab we learned how to find specific values and addresses of different registers and memories. Additionally, we learned how to manipulate these values for future labs. Furthermore, learning how to debug future assembly code by setting break points and looking at certain values were some of the biggest and most useful things learned in this lab.